Synopsis

Behavioral Segmentation for Targeted Marketing

Problem Id: 10

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Problem Description:

- In the modern retail landscape, "one-size-fits-all" marketing is inefficient. Businesses face the challenge of understanding diverse customer behaviors to optimize their promotional strategies. This project addresses the need for Behavioral Segmentation, where customers are grouped based on their actual interactions with a brand—specifically their Annual Spending and Purchase Frequency

- By identifying distinct customer "personas," such as "high-spenders" or "frequent browsers," companies can tailor their marketing efforts, improve customer retention, and maximize Return on Investment (ROI). This mirrors real-life applications in e-commerce and retail banking, where machine learning automates the discovery of hidden patterns in massive datasets to drive personalized consumer experiences.

- **Steps of Implementation:**

This section should explain the each phase of ML pipeline used.

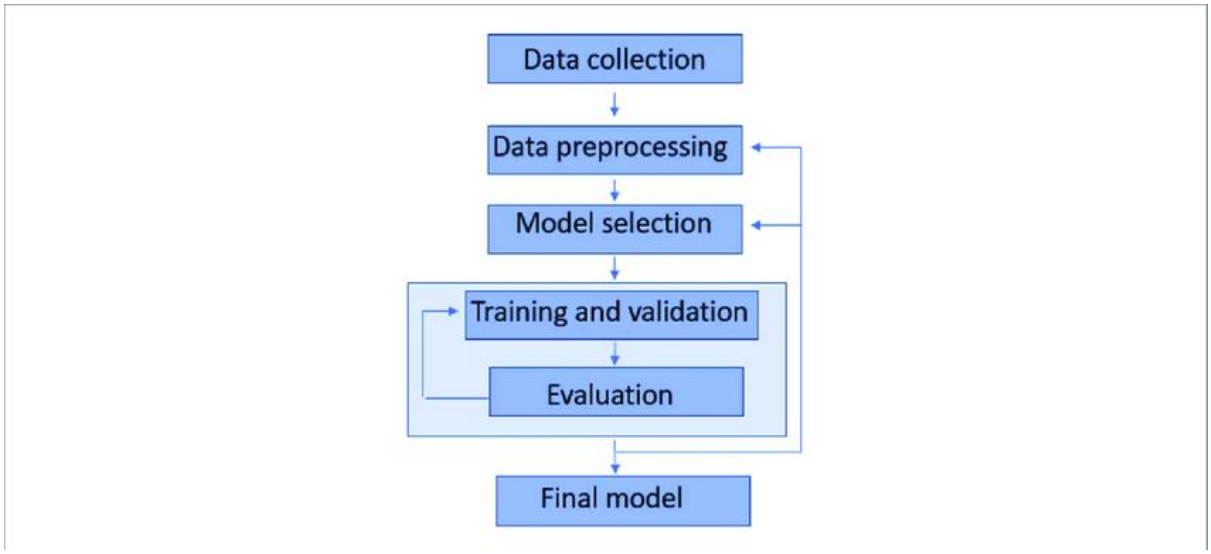

Fig. 1 Sample architecture

• Expected Modules/ Libraries to be used :

- **Pandas:** Used for data manipulation and creating the initial 150-row retail dataframe.
- **NumPy:** Essential for numerical computations and generating the synthetic data distributions.
- **Scikit-learn (Sklearn):** Provides the KMean algorithm and StandardScaler for data normalization.
- **Matplotlib/Seaborn:** Used for data visualization, specifically for the Elbow Method graph and the final cluster scatter plots.

- # Expected Evaluation metrics and Plots to be used:

- **The Elbow Method Plot:** A line graph showing the relationship between the number of clusters and WCSS. We look for the "elbow" point where the rate of decrease shifts, indicating the optimal K.

- **WCSS (Within-Cluster Sum of Squares):** A metric used to evaluate how compact the clusters are; lower values generally indicate tighter groupings.

- **Cluster Scatter Plot:** A 2D visualization of Annual_Spending vs. Purchase_Frequency, color-coded by cluster, to qualitatively validate the "personas" identified by the model.

- # Required topics from the subject:

- **Strings** – Used for handling categorical labels (“High Risk”, “Low Risk”) before encoding.

- **File Handling** – Reading/writing CSV files for storing synthetic health records and saving evaluation results.

- **Lists and Dictionaries:** Used to store the synthetic data before converting it into a structured format.

- **Classes and Objects:** Since Scikit-learn uses an Object-Oriented approach, you must understand how to "instantiate" a model (e.g., kmeans = KMeans()) and call methods like .fit() and .predict().

- **Functions:** Helpful for wrapping the "Retail Data Synthesis" process into a r eusable block of code.

- **Loops (For/While):** Specifically required for the **Elbow Method**, where you must loop through different values of $ (e.g., 1 to 10) to calculate the WCSS for each.

- # Platform Required:

- **Jupyter Notebook** – For interactive coding, visualization, and documentation.

- **Python 3.x -** The core programming language for the implementation.

- # Books and Link Sources:

• *Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow* by Aurélien Géron.

• Campus x Youtube channel-https://youtu.be/Zja1aZXrlDE?si=a8Y_8AGZcda5uXUc

- Scikit-learn Official Documentation: `sklearn.cluster.KMeans`.

- Kaggle Tutorial: "Customer Segmentation using K-Means Clustering".